

IP-SAKTI SAHAYAK (■■■-■■■■■■■ ■■■■■■■■)

AI-Powered Intellectual Property & Traditional Knowledge Intelligence Platform | SIH 2026 — PS26045

Platform Classification: Specialized Regulatory & Legal Intelligence System | **Target Domain:** Indian Patents Act 1970, Ayurvedic Formulary (TKDL/AYUSH), Biological Diversity Act 2002 & 2023, International Treaties (PCT/WIPO/TRIPS) | **Core Philosophy:** Citation-Grounded RAG with Safe Abstention (Zero Hallucination Tolerance).

1. Executive Summary & Problem Context

The Problem: Innovators, research scholars, and enterprises dealing with Ayurveda, natural products, and traditional knowledge face a heavily fragmented regulatory maze in India and globally. Critical regulations are governed by disconnected statutory bodies: the **Indian Patent Office (IP India)**, the **Ministry of AYUSH** (Drugs & Cosmetics Act & Rules), the **National Biodiversity Authority (NBA)** (Biological Diversity Act 2002 & 2023 Amendments), the **Food Safety and Standards Authority of India (FSSAI)** (Ayurveda Aahara), the **Traditional Knowledge Digital Library (TKDL)**, and international entities (**WIPO, PCT, WTO TRIPS**). General-purpose LLMs (ChatGPT, Gemini, Claude) frequently hallucinate statutory sections, invent case law, and misapply patentability exceptions, causing catastrophic legal risk.

The Solution: **IP-SAKTI SAHAYAK** is an authoritative, citation-backed legal intelligence platform engineered specifically for SIH 2026 (Problem Statement PS26045). It operates on a dual principle of *strict statutory evidence retrieval* and *deterministic safe abstention*. Every output is tied to audited citations [Source N], claims are traced back to exact provisions, and when evidence is incomplete or queries fall outside the law, the system deliberately refuses to speculate, escalating to human legal experts.

2. Architectural Design & 6-Stage Deterministic RAG Pipeline

The website is backed by a sophisticated multi-stage pipeline ensuring end-to-end evidence verifiability before any response is rendered to the user:

Stage	Subsystem	Technical Mechanism & Guarantees
01. ASK	Question Intake & Normalization	Multi-language query intake (English, Hindi, Odia). Locks statutory legal proper nouns (e.g. <i>Section 3(d)</i> , <i>Rule 158B</i> , <i>ABS</i> , <i>TRIPS</i> , <i>TKDL</i>) to prevent mistranslation or context drifting.
02. CLASSIFY	Domain & Jurisdiction Dispatch	Determines legal domain (Patents, Trademarks, AYUSH, Biodiversity, GI) and target regulator (IP India, NBA, FSSAI, CDSCO, WIPO), preventing cross-jurisdictional legal pollution.
03. RETRIEVE	Hybrid Dual-Channel Search	Merges Dense Semantic Retrieval (FAISS vector store powered by multilingual BAAI/bge-m3 embeddings) and Sparse Keyword Search (Rank-BM25), with auto-tuned boosts for statutory section numbers.
04. VERIFY	Freshness & Cross-Encoder Rerank	Filters top 20 candidates down to top 5 evidence chunks using a cross-encoder reranker. Verifies legal freshness and amendment status, excluding superseded gazettes.
05. EXPLAIN	Grounded LLM Generation	Groq-hosted high-speed LLM generates responses under strict temperature constraints. Every single factual statement is mapped to a mandatory [Source N] audit badge with full claim traceability.
06. ESCALATE	Safe Abstention Gatekeeper	Calculates a composite confidence score (0.00 – 1.00). If confidence is below threshold (< 0.40) or evidence conflicts, the system safely abstains, highlights missing legal parameters, and queues expert review.

3. Comprehensive Breakdown of Website Modules & Pages

The IP-SAKTI Sahayak platform is organized into 14 distinct specialized interfaces, accessible seamlessly via the navigation bar and central dashboard:

Flagship: Innovation & Patentability Analyzer (/analyze-innovation)

The platform's premier research tool. Users enter an invention description (e.g. herbal extract, formulation, process). The engine performs: (1) Botanical & phytochemical entity extraction (mapping to classical Sanskrit binomials); (2) Traditional Knowledge & TKDL overlap check; (3) Indian Patentability test under Sections 3(d), 3(p), 3(e), 3(h), 3(j); (4) Regulatory manufacturing pathway (AYUSH Rule 158B, Schedule T GMP); (5) Biodiversity/NBA Access & Benefit Sharing (ABS) requirements; (6) 5-Axis IP Risk Radar computation; and (7) Instant downloadable PDF/HTML audit dossiers.

AI Legal Assistant / Citation-Backed Chat (/chat)

Interactive legal conversational interface equipped with dual operating modes: (A) Quick Hybrid RAG for immediate citation-grounded statutory answers; and (B) 5-Track Agentic Deep Research for comprehensive legal opinions. Features claim-level 'Why did AI say this?' evidence traceability modals and built-in voice bot listening/speaking capabilities.

Regulatory RAG Guidance Engine (/regulatory-rag)

Dedicated statutory subsystem divided into 4 tabbed workspaces: (1) Statutory Query Engine (filtered by jurisdiction, regulator, and legal regime); (2) Unified Multi-Router (routing between Patents, AYUSH, Food, and Biodiversity); (3) Registered Regulatory Authorities directory; and (4) Live Amendment Timelines tracking official gazettes.

Prior-Art & TKDL Explorer (/prior-art)

Specialized search engine cross-referencing modern invention disclosures with ancient Ayurvedic texts (Charaka Samhita, Sushruta Samhita, Astanga Hridaya, Bhavaprakasha, API) and international patent registries. Evaluates concept differentiation and prior statutory disclosure to defeat anticipation rejections.

ABS & Biodiversity Compliance Engine (/compliance)

Direct compliance portal for the Biological Diversity Act 2002 and the 2023 Statutory Amendments. Automatically determines whether foreign or domestic entities require NBA Form I, Form II, or Form III approvals, State Biodiversity Board (SBB) intimations, or qualify for Section 40 commercial exemptions.

Comparative Jurisdiction Matrix (/comparative)

Side-by-side legal comparative intelligence matrix benchmarking India against the US (USPTO), Europe (EPO), and International bodies (WIPO/PCT). Details disparate stances on natural products, Section 3(d) therapeutic efficacy standards, Alice/Myriad diagnostic exceptions, and Nagoya Protocol compliance.

Regulation Time Machine & Gazette Monitor (/regulations)

Interactive legislative tracker maintaining chronological gazette records. Features an interactive 'What Changed?' side-by-side textual diff comparing prior statutes (e.g. Biological Diversity Act 2002 vs. 2023 Amendment; Patent Rules 2003 vs. 2024 Rules).

Interactive IP & TK Knowledge Graph (/knowledge-graph)

Dynamic, force-directed network visualizer displaying interconnected nodes across Botanicals, Phytochemicals, Classical Formulations, Ayurvedic Treatises, Statutory Sections, and Regulatory Authorities. Enables multi-hop relational discovery for patent examiners and litigators.

Document Intelligence & Extraction (/documents)

Automated ingestion portal for PDF and DOCX legal instruments (agreements, patent specifications, ABS contracts). Automatically parses and extracts defined clauses, cited statutory sections, governing authorities, filing deadlines, and compliance obligations.

Live IP Analytics & Patents Dashboard (/ip-analytics, /patents-dashboard)

Executive analytics suite visualizing Indian IP trends, AYUSH patent filing velocities, grant versus rejection ratios under Section 3(p), CSIR traditional knowledge revocation histories, and geographic patent densities.

Evaluation Benchmark & Red-Team Defense (/benchmark)

Public verification and auditing dashboard testing the RAG engine against standardized evaluation metrics (Recall@K, Groundedness %, Context Relevance) and adversarial red-team trap queries designed to catch un-grounded hallucinations.

Authoritative Sources Registry (/sources)

Provenance registry listing indexed official databases: IP India, India Code, Ministry of AYUSH, NBA, TKDL, WIPO Lex, WTO TRIPS, and WHO. Details authority tiers, update frequencies, and official gazette URLs.

Personal Profile, Referral Program & User Auth (/dashboard, /login, /signup)

Full user identity management enabling persistent session states, personalized case dossiers, and an exclusive Referral Desk generating unique researcher invite links (e.g. REF-USERNAME-XXXX) to earn research credits.

Administrative Diagnostics (/admin)

System monitoring panel displaying vector store memory, BM25 indices, database connection pools, cache hit ratios, and background pipeline health.

4. Technology Stack & Architectural Highlights

Component	Technology / Framework	Role & Implementation Highlights
Backend Framework	Python 3.10+, Flask 3.0.3	Modular Blueprint architecture (routes, services, AI, RAG, regulatory).
LLM Engine	Groq API (OpenAI-compatible)	High-throughput, low-latency grounded statutory synthesis with claim tagging.
Embeddings	BAAI/bge-m3 (HuggingFace)	1024-dim multilingual dense embeddings for cross-lingual English-Hindi-Odia matching.
Vector Store	FAISS (faiss-cpu 1.8.0)	Fast L2 similarity index over statutory sections and gazette passages.
Sparse Search	Rank-BM25 (rank_bm25)	Keyword scoring fine-tuned for statutory section numbers and botanical names.
Reranking	Cross-Encoder (sentence-transformers)	Deep semantic reranking of candidate chunks to distill top 5 evidence paragraphs.
Relational DB	SQLite + SQLAlchemy 2.0	User authentication, referral codes, search history, and metadata registry.
Document Parsing	PyMuPDF (fitz) & PyPDF	High-fidelity extraction of legal acts, official gazettes, and uploaded specs.
Dossier Reports	ReportLab 4.2.5	Dynamic generation of official audit certificates, PDF dossiers, and summary briefs.
Frontend UI	HTML5, CSS3, Vanilla JS	Custom design system with zero Tailwind dependency, glassmorphism, and responsive cards.

5. Key Differentiators & Trust Guarantees

- **Zero Hallucination Tolerance:** General LLMs guess legal outcomes; IP-SAKTI Sahayak strictly enforces evidence grounding and citation matching before presenting text.
- **Safe Abstention as a Feature:** The system deliberately abstains when evidence is missing, conflicting, or out-of-domain, avoiding legal liabilities.
- **Multilingual Equity (EN / HI / OR):** Democratizes complex IP and patent law for grassroots Ayurvedic practitioners and MSME innovators in vernacular languages.
- **Graceful Subsystem Degradation:** If HuggingFace or vector models are inaccessible, the engine smoothly transitions to lightweight keyword mode without service disruption.
- **Strict Data Privacy & Security:** No third-party data tracking, sanitization of file uploads, and session-guarded referral profile management.

Statutory Disclaimer: IP-SAKTI SAHAYAK is an artificial intelligence-powered regulatory guidance and document research tool developed for informational and academic evaluation under Smart India Hackathon 2026. It does not constitute formal legal counsel or an official patent examination opinion. Users are advised to cross-verify all guidance with the Controller General of Patents, Designs and Trade Marks (IP India), Ministry of AYUSH, or a certified patent attorney.